

HairOS

Dermatologist Blind Review Pack

Version 1.0.0 · 25 cases · Pack date: 2026-06-06

Reviewer instructions

Thank you for participating in the HairOS clinical validation. This pack contains 25 patient cases distilled to the patient- and doctor-facing outputs that the system would surface in production. No internal scoring, registries, or implementation details are included by design.

Please read each case as if it had been generated for a patient under your care. For every case, complete the matching entry in the scoring form. Use 1–5 Likert scoring (1 = unacceptable, 5 = excellent). Categorise every negative comment using the categories on the form. Conclude each case with a sign decision: YES, YES WITH MINOR EDITS, or NO.

Reviewer eligibility, gate thresholds, and aggregation methodology are documented in CLINICAL_ACCEPTANCE_GATE_SPEC.md (provided separately). Submit your completed form through the anonymised intake channel within 21 days.

Pack composition

- Male Androgenetic Alopecia: 5 cases
- Female Pattern Hair Loss: 4 cases
- PCOS-Associated Hair Loss: 4 cases
- Acute Telogen Effluvium: 3 cases
- Chronic Telogen Effluvium: 2 cases
- Alopecia Areata Spectrum: 3 cases
- Inflammatory Scalp Disorder: 2 cases
- Multifactorial Hair Loss: 2 cases

Total: 25 cases spanning mild, moderate, severe, ambiguous, and overlapping presentations.

Case RC-01 — Male Androgenetic Alopecia

Case 1 of 25

Patient Questionnaire

Sex: Male

Age group: young adult

Duration of concern: 3–6 months

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 1

Hair changes: Receding hairline

Scalp state: Normal scalp

Self-reported causes: None

Lifestyle factors: Chronic stress; Poor sleep

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: Iron deficiency

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Male
- Young adult (< 25)
- Iron / ferritin deficiency reported
- Chronic stress / poor sleep

Detected Drivers

- Follicular miniaturization — Primary driver
- Nutritional substrate limitation — Supporting driver
- Telogen cycle disruption — Modulator

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Mild

Co-explanations: Nutritional Hair Loss

Root Cause Explanation

Male, age 19, with progressive Norwood-1 pattern thinning concentrated at crown/temporal regions, and minimal active shedding. The pattern topology (terminal-to-vellus transition without diffuse shed) is the definitional signature of androgen-driven miniaturization. Pathway activation is dominated by follicular miniaturization (e0.60) with the telogen pathway only modestly activated, which the cause registry's log-likelihood weights resolve into a dominant posterior for androgen-driven miniaturization. The case therefore leads on the androgenic cause with the standard MPHL protocol class and Norwood-grade-locked kit gating.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Scalp inflammation control
- Nutritional repletion

Male AGA pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Scalp inflammation control, Nutritional repletion.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months
- Ferritin — repeat at 4 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol. We are also addressing the contributing role of Nutritional Hair Loss.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Co-explanations: Nutritional Hair Loss. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-02 — Male Androgenetic Alopecia

Case 2 of 25

Patient Questionnaire

Sex: Male

Age group: adult

Duration of concern: 6–12 months

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 3

Hair changes: Receding hairline

Scalp state: Normal scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Male
- Adult (25–39)
- Family history of hair loss

Detected Drivers

- Follicular miniaturization — Primary driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Moderate

Root Cause Explanation

Male, age 27, with progressive Norwood-3 pattern thinning concentrated at crown/temporal regions, family history positive, and minimal active shedding. The pattern topology (terminal-to-vellus transition without diffuse shed) is the definitional signature of androgen-driven miniaturization. Pathway activation is dominated by follicular miniaturization ("e0.60) with the telogen pathway only modestly activated, which the cause registry's log-likelihood weights resolve into a dominant posterior for androgen-driven miniaturization. The case therefore leads on the androgenic cause with the standard MPHL protocol class and Norwood-grade-locked kit gating.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus

transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Scalp inflammation control

Male AGA pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon.

Case RC-03 — Male Androgenetic Alopecia

Case 3 of 25

Patient Questionnaire

Sex: Male

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 5

Hair changes: Thinning at crown; Receding hairline

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: Chronic stress; Poor sleep

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Advanced-grade severity (4–5)
- Male
- Adult (25–39)
- Family history of hair loss
- Oily scalp
- Chronic stress / poor sleep

Detected Drivers

- Follicular miniaturization — Primary driver
- Telogen cycle disruption — Modulator

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Severe

Root Cause Explanation

Male, age 35, with progressive Norwood-5 pattern thinning concentrated at crown/temporal regions, family history positive, seborrheic scalp tendency, and minimal active shedding. The pattern topology (terminal-to-vellus transition without diffuse shed) is the definitional signature of androgen-driven miniaturization. Pathway activation is dominated by follicular miniaturization ("e0.60) with the telogen pathway only modestly activated, which the cause registry's log-likelihood weights resolve into a dominant posterior for androgen-driven miniaturization. The case therefore leads on the androgenic cause with the standard MPHL protocol class and Norwood-grade-locked kit gating.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Scalp inflammation control

Male AGA pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Severity warrants combination therapy and an early dermatologist follow-up at 12 weeks. Therapy axes engaged: DHT suppression, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon.

Case RC-04 — Male Androgenetic Alopecia

Case 4 of 25

Patient Questionnaire

Sex: Male

Age group: mature adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 6

Hair changes: Thinning at crown; Receding hairline

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: Iron deficiency

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Advanced-grade severity (4–5)
- Male
- Mature adult (40–54)
- Family history of hair loss
- Oily scalp
- Iron / ferritin deficiency reported

Detected Drivers

- Follicular miniaturization — Primary driver
- Nutritional substrate limitation — Supporting driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Severe

Co-explanations: Nutritional Hair Loss

Root Cause Explanation

Male, age 42, with progressive Norwood-6 pattern thinning concentrated at crown/temporal regions, family history positive, seborrheic scalp tendency, and minimal active shedding. The pattern topology (terminal-to-vellus transition without diffuse shed) is the definitional signature of androgen-driven miniaturization. Pathway activation is dominated by follicular miniaturization ("e0.60) with the telogen pathway only modestly activated, which the cause registry's log-likelihood weights resolve into a dominant posterior for androgen-driven miniaturization. The case therefore leads on the androgenic cause with the standard MPHL protocol class and Norwood-grade-locked kit gating.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Scalp inflammation control
- Nutritional repletion

Male AGA pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Severity warrants combination therapy and an early dermatologist follow-up at 12 weeks. Therapy axes engaged: DHT suppression, Scalp inflammation control, Nutritional repletion.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months
- Ferritin — repeat at 4 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol. We are also addressing the contributing role of Nutritional Hair Loss.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Co-explanations: Nutritional Hair Loss. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-05 — Male Androgenetic Alopecia

Case 5 of 25

Patient Questionnaire

Sex: Male

Age group: mature adult

Duration of concern: 6–12 months

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 2

Hair changes: Receding hairline

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Male
- Mature adult (40–54)
- Family history of hair loss
- Oily scalp

Detected Drivers

- Follicular miniaturization — Primary driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Mild

Root Cause Explanation

Male, age 51, with progressive Norwood-2 pattern thinning concentrated at crown/temporal regions, family history positive, seborrheic scalp tendency, and minimal active shedding. The pattern topology (terminal-to-vellus transition without diffuse shed) is the definitional signature of androgen-driven miniaturization. Pathway activation is dominated by follicular miniaturization ("e0.60) with the telogen pathway only modestly activated, which the cause registry's log-likelihood weights resolve into a dominant posterior for androgen-driven miniaturization. The case therefore leads on the androgenic cause with the standard MPHL protocol class and Norwood-grade-locked kit gating.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive

cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Scalp inflammation control

Male AGA pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Conflicting signals present — multifactorial framing preferred.

Case RC-06 — Female Pattern Hair Loss

Case 6 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 2

Hair changes: Widening parting; Thinning at crown

Scalp state: Normal scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: Iron deficiency

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Female
- Adult (25–39)
- Family history of hair loss
- Iron / ferritin deficiency reported

Detected Drivers

- Follicular miniaturization — Primary driver
- Endocrine dysregulation — Supporting driver
- Nutritional substrate limitation — Supporting driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Mild

Co-explanations: Endocrine-Driven Hair Loss; Nutritional Hair Loss

Root Cause Explanation

Female, age 28, Ludwig 1 pattern with widening parting, family history positive, minimal active shedding. The crown-emphasized topology with intact frontal hairline is the female-pattern signature of androgen-driven miniaturization expressed within an endocrine context. Both miniaturization ("e0.55) and hormonal-dysregulation ("e0.40) pathways activate, producing a co-leading androgenic + hormonal cause posterior — the legacy DiagnosisKey for FPHL maps directly to this co-explanation.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Endocrine optimisation
- Scalp inflammation control
- Nutritional repletion

FPHL pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Endocrine optimisation, Scalp inflammation control, Nutritional repletion.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months
- Ferritin — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol. We are also addressing the contributing role of Endocrine-Driven Hair Loss and Nutritional Hair Loss.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Co-explanations: Endocrine-Driven Hair Loss, Nutritional Hair Loss. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-07 — Female Pattern Hair Loss

Case 7 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 3

Hair changes: Widening parting; Thinning at crown

Scalp state: Normal scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Female
- Adult (25–39)
- Family history of hair loss

Detected Drivers

- Follicular miniaturization — Primary driver
- Endocrine dysregulation — Supporting driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Moderate

Co-explanations: Endocrine-Driven Hair Loss

Root Cause Explanation

Female, age 36, Ludwig 2 pattern with widening parting, family history positive, minimal active shedding. The crown-emphasized topology with intact frontal hairline is the female-pattern signature of androgen-driven miniaturization expressed within an endocrine context. Both miniaturization ("e0.55) and hormonal-dysregulation ("e0.40) pathways activate, producing a co-leading androgenic + hormonal cause posterior — the legacy DiagnosisKey for FPHL maps directly to this co-explanation.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus

transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Endocrine optimisation
- Scalp inflammation control

FPHL pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Endocrine optimisation, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol. We are also addressing the contributing role of Endocrine-Driven Hair Loss.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Co-explanations: Endocrine-Driven Hair Loss.

Case RC-08 — Female Pattern Hair Loss

Case 8 of 25

Patient Questionnaire

Sex: Female

Age group: mature adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 2

Hair changes: Widening parting; Thinning at crown

Scalp state: Normal scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: Perimenopause

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Female
- Mature adult (40–54)
- Family history of hair loss
- Perimenopausal

Detected Drivers

- Follicular miniaturization — Primary driver
- Endocrine dysregulation — Supporting driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Mild

Co-explanations: Endocrine-Driven Hair Loss

Root Cause Explanation

Female, age 52, Ludwig 1 pattern with widening parting, perimenopausal endocrine state, family history positive, minimal active shedding. The crown-emphasized topology with intact frontal hairline is the female-pattern signature of androgen-driven miniaturization expressed within an endocrine context. Both miniaturization ("e0.55) and hormonal-dysregulation ("e0.40) pathways activate, producing a co-leading androgenic + hormonal cause posterior — the legacy DiagnosisKey for FPHL maps directly to this co-explanation.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Endocrine optimisation
- Scalp inflammation control

FPHL pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Endocrine optimisation, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol. We are also addressing the contributing role of Endocrine-Driven Hair Loss.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Co-explanations: Endocrine-Driven Hair Loss. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-09 — Female Pattern Hair Loss

Case 9 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible, minimal fall

Reported grade: Grade 2

Hair changes: Widening parting; Thinning at crown

Scalp state: Normal scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Thinning without active shedding
- Early-grade severity (1–3)
- Female
- Adult (25–39)
- Family history of hair loss

Detected Drivers

- Follicular miniaturization — Primary driver
- Endocrine dysregulation — Supporting driver

Diagnosis

Primary diagnosis: Androgenetic / Pattern Hair Loss

Severity: Mild

Co-explanations: Endocrine-Driven Hair Loss

Root Cause Explanation

Female, age 32, Ludwig 1 pattern with widening parting, family history positive, minimal active shedding. The crown-emphasized topology with intact frontal hairline is the female-pattern signature of androgen-driven miniaturization expressed within an endocrine context. Both miniaturization ("e0.55) and hormonal-dysregulation ("e0.40) pathways activate, producing a co-leading androgenic + hormonal cause posterior — the legacy DiagnosisKey for FPHL maps directly to this co-explanation.

Biological Mechanism

DHT-driven dermal-papilla signalling shortens anagen and reduces follicle volume across successive cycles. Each cycle produces a progressively finer, shorter, and lighter hair until terminal-to-vellus

transformation. Follicular architecture is preserved early and degrades with sustained exposure; partial reversibility is possible with timely intervention.

Protocol Recommendation

- DHT suppression
- Endocrine optimisation
- Scalp inflammation control

FPHL pattern protocol focused on DHT suppression and scalp inflammation control, with a maintenance horizon. Initiated at standard intensity with 12-week reassessment. Therapy axes engaged: DHT suppression, Endocrine optimisation, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Shedding stabilises within 8–12 weeks. Caliber improvement visible from 16–24 weeks. Density change visible from 6–9 months. Maintenance is lifelong; discontinuation typically reverts progress within 6–12 months.

Patient Narrative

Your hair is getting finer over time because of how your follicles respond to hormones — this is a hereditary, pattern-based form of hair loss. Treatment slows the pattern and supports density; results build over months, not weeks, and improvement requires staying on the protocol. We are also addressing the contributing role of Endocrine-Driven Hair Loss.

Doctor Narrative

Pattern thinning topology with intact follicular architecture; androgen-driven miniaturization is the dominant explanation. Protocol focuses on DHT suppression and scalp inflammation control, with a maintenance horizon. Co-explanations: Endocrine-Driven Hair Loss.

Case RC-10 — PCOS-Associated Hair Loss

Case 10 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible with moderate fall

Reported grade: Grade 2

Hair changes: Widening parting; Thinning at crown

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: PCOS; Irregular periods

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- PCOS
- Female
- Adult (25–39)
- Oily scalp
- Early-grade severity (1–3)
- PCOS with metabolic features

Detected Drivers

- Endocrine dysregulation — Primary driver
- Follicular miniaturization — Supporting driver
- Metabolic dysfunction — Supporting driver

Diagnosis

Primary diagnosis: Endocrine-Driven Hair Loss

Severity: Mild

Co-explanations: Androgenetic / Pattern Hair Loss; Metabolic-Driven Hair Loss

Root Cause Explanation

26-year-old woman with PCOS diagnosis and metabolic features, oily scalp, pattern thinning at crown, Grade 2 severity, family history positive. The endocrine driver (PCOS) carries the strongest log-likelihood weight in the cause registry for this signal mix; miniaturization activates as a downstream mechanism but the named driver is hormonal because endocrine signals dominate. Metabolic-dysfunction pathway co-activates via insulin–androgen cross-talk. Protocol class is PCOS with mandatory metabolic and endocrine monitoring.

Biological Mechanism

Perturbation of the thyroid, androgen, or estrogen axes modulates follicle cycling. Hypothyroid states prolong telogen; androgen excess (PCOS) drives miniaturization via follicular sensitivity to DHT; declining estrogen at peri/menopause withdraws an anagen-protective signal. Phenotype tracks the dominant axis.

Protocol Recommendation

- Endocrine optimisation
- DHT suppression
- Scalp inflammation control
- Metabolic optimisation

Endocrine optimisation upstream of hair therapy; hair-directed components run in parallel but the named lever is endocrine. Therapy axes engaged: Endocrine optimisation, DHT suppression, Scalp inflammation control, Metabolic optimisation.

Monitoring Plan

Required

- Global photography — every 3 months
- Androgen panel — repeat at 6 months
- Menstrual cycle diary
- HbA1c — repeat at 3 months
- Weight tracking — monthly

Recommended

- Shed count — monthly diary

Expected Outcomes

Hair response lags endocrine correction by 8–12 weeks. Full expected response window is 6–9 months from endocrine stabilisation.

Patient Narrative

A hormonal shift is the leading explanation for what you're seeing. Addressing the underlying endocrine state, alongside targeted hair therapy, gives the best chance of improvement over the next 6–9 months. We are also addressing the contributing role of Androgenetic / Pattern Hair Loss and Metabolic-Driven Hair Loss.

Doctor Narrative

Identifiable endocrine perturbation is the named driver. Correcting the upstream endocrine state typically improves the hair phenotype on standard cycles; hair-directed therapy runs in parallel. Co-explanations: Androgenetic / Pattern Hair Loss, Metabolic-Driven Hair Loss.

Case RC-11 — PCOS-Associated Hair Loss

Case 11 of 25

Patient Questionnaire

Sex: Female

Age group: young adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible with moderate fall

Reported grade: Grade 3

Hair changes: Widening parting; Thinning at crown

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: PCOS

Immunity / autoimmune: None

Reported deficiencies: Iron deficiency; Vitamin B12 deficiency

Gut symptoms: Bloating

Diet pattern: Post-bariatric

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- PCOS
- Female
- Young adult (< 25)
- Oily scalp
- Early-grade severity (1–3)
- Iron / ferritin deficiency reported
- Vitamin B12 deficiency reported
- Persistent bloating

Detected Drivers

- Endocrine dysregulation — Primary driver
- Nutritional substrate limitation — Supporting driver
- Follicular miniaturization — Supporting driver

Diagnosis

Primary diagnosis: Endocrine-Driven Hair Loss

Severity: Moderate

Co-explanations: Androgenetic / Pattern Hair Loss; Nutritional Hair Loss

Root Cause Explanation

19-year-old woman with PCOS diagnosis, oily scalp, pattern thinning at crown, Grade 3 severity, family history positive. The endocrine driver (PCOS) carries the strongest log-likelihood weight in the cause registry for this signal mix; miniaturization activates as a downstream mechanism but the named driver is hormonal because endocrine signals dominate. Post-bariatric nutritional state co-activates nutritional-limitation pathway requiring parallel repletion. Protocol class is PCOS with mandatory metabolic and

endocrine monitoring.

Biological Mechanism

Perturbation of the thyroid, androgen, or estrogen axes modulates follicle cycling. Hypothyroid states prolong telogen; androgen excess (PCOS) drives miniaturization via follicular sensitivity to DHT; declining estrogen at peri/menopause withdraws an anagen-protective signal. Phenotype tracks the dominant axis.

Protocol Recommendation

- Endocrine optimisation
- DHT suppression
- Scalp inflammation control
- Nutritional repletion
- Gut repair / absorption support

Endocrine optimisation upstream of hair therapy; hair-directed components run in parallel but the named lever is endocrine. Therapy axes engaged: Endocrine optimisation, DHT suppression, Scalp inflammation control, Nutritional repletion, Gut repair / absorption support.

Monitoring Plan

Required

- Global photography — every 3 months
- Androgen panel — repeat at 6 months
- Menstrual cycle diary
- Ferritin — repeat at 4 months
- Vitamin B12 — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months

Recommended

- Shed count — monthly diary

Expected Outcomes

Hair response lags endocrine correction by 8–12 weeks. Full expected response window is 6–9 months from endocrine stabilisation.

Patient Narrative

A hormonal shift is the leading explanation for what you're seeing. Addressing the underlying endocrine state, alongside targeted hair therapy, gives the best chance of improvement over the next 6–9 months. We are also addressing the contributing role of Androgenetic / Pattern Hair Loss and Nutritional Hair Loss.

Doctor Narrative

Identifiable endocrine perturbation is the named driver. Correcting the upstream endocrine state typically improves the hair phenotype on standard cycles; hair-directed therapy runs in parallel. Co-explanations: Androgenetic / Pattern Hair Loss, Nutritional Hair Loss. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-12 — PCOS-Associated Hair Loss

Case 12 of 25

Patient Questionnaire

Sex: Female

Age group: young adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible with moderate fall

Reported grade: Grade 4

Hair changes: Widening parting; Thinning at crown

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: PCOS; Irregular periods

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- PCOS
- Female
- Young adult (< 25)
- Oily scalp
- Advanced-grade severity (4–5)
- PCOS with metabolic features

Detected Drivers

- Endocrine dysregulation — Primary driver
- Follicular miniaturization — Supporting driver
- Metabolic dysfunction — Supporting driver

Diagnosis

Primary diagnosis: Endocrine-Driven Hair Loss

Severity: Severe

Co-explanations: Androgenetic / Pattern Hair Loss; Metabolic-Driven Hair Loss

Root Cause Explanation

24-year-old woman with PCOS diagnosis and metabolic features, oily scalp, pattern thinning at crown, Grade 4-5 severity, family history positive. The endocrine driver (PCOS) carries the strongest log-likelihood weight in the cause registry for this signal mix; miniaturization activates as a downstream mechanism but the named driver is hormonal because endocrine signals dominate. Metabolic-dysfunction pathway co-activates via insulin–androgen cross-talk. Protocol class is PCOS with mandatory metabolic and endocrine monitoring.

Biological Mechanism

Perturbation of the thyroid, androgen, or estrogen axes modulates follicle cycling. Hypothyroid states prolong telogen; androgen excess (PCOS) drives miniaturization via follicular sensitivity to DHT; declining estrogen at peri/menopause withdraws an anagen-protective signal. Phenotype tracks the dominant axis.

Protocol Recommendation

- Endocrine optimisation
- DHT suppression
- Scalp inflammation control
- Metabolic optimisation

Endocrine optimisation upstream of hair therapy; hair-directed components run in parallel but the named lever is endocrine. Therapy axes engaged: Endocrine optimisation, DHT suppression, Scalp inflammation control, Metabolic optimisation.

Monitoring Plan

Required

- Global photography — every 3 months
- Androgen panel — repeat at 6 months
- Menstrual cycle diary
- HbA1c — repeat at 3 months
- Weight tracking — monthly

Recommended

- Shed count — monthly diary

Expected Outcomes

Hair response lags endocrine correction by 8–12 weeks. Full expected response window is 6–9 months from endocrine stabilisation.

Patient Narrative

A hormonal shift is the leading explanation for what you're seeing. Addressing the underlying endocrine state, alongside targeted hair therapy, gives the best chance of improvement over the next 6–9 months. We are also addressing the contributing role of Androgenetic / Pattern Hair Loss and Metabolic-Driven Hair Loss.

Doctor Narrative

Identifiable endocrine perturbation is the named driver. Correcting the upstream endocrine state typically improves the hair phenotype on standard cycles; hair-directed therapy runs in parallel. Co-explanations: Androgenetic / Pattern Hair Loss, Metabolic-Driven Hair Loss.

Case RC-13 — PCOS-Associated Hair Loss

Case 13 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: thinning visible with moderate fall

Reported grade: Grade 3

Hair changes: Widening parting; Thinning at crown

Scalp state: Oily scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: PCOS

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- PCOS
- Female
- Adult (25–39)
- Oily scalp
- Early-grade severity (1–3)

Detected Drivers

- Endocrine dysregulation — Primary driver
- Follicular miniaturization — Supporting driver

Diagnosis

Primary diagnosis: Endocrine-Driven Hair Loss

Severity: Moderate

Co-explanations: Androgenetic / Pattern Hair Loss

Root Cause Explanation

30-year-old woman with PCOS diagnosis, oily scalp, pattern thinning at crown, Grade 3 severity, family history positive. The endocrine driver (PCOS) carries the strongest log-likelihood weight in the cause registry for this signal mix; miniaturization activates as a downstream mechanism but the named driver is hormonal because endocrine signals dominate. Protocol class is PCOS with mandatory metabolic and endocrine monitoring.

Biological Mechanism

Perturbation of the thyroid, androgen, or estrogen axes modulates follicle cycling. Hypothyroid states prolong telogen; androgen excess (PCOS) drives miniaturization via follicular sensitivity to DHT;

declining estrogen at peri/menopause withdraws an anagen-protective signal. Phenotype tracks the dominant axis.

Protocol Recommendation

- Endocrine optimisation
- DHT suppression
- Scalp inflammation control

Endocrine optimisation upstream of hair therapy; hair-directed components run in parallel but the named lever is endocrine. Therapy axes engaged: Endocrine optimisation, DHT suppression, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Androgen panel — repeat at 6 months
- Menstrual cycle diary

Recommended

- Shed count — monthly diary

Expected Outcomes

Hair response lags endocrine correction by 8–12 weeks. Full expected response window is 6–9 months from endocrine stabilisation.

Patient Narrative

A hormonal shift is the leading explanation for what you're seeing. Addressing the underlying endocrine state, alongside targeted hair therapy, gives the best chance of improvement over the next 6–9 months. We are also addressing the contributing role of Androgenetic / Pattern Hair Loss.

Doctor Narrative

Identifiable endocrine perturbation is the named driver. Correcting the upstream endocrine state typically improves the hair phenotype on standard cycles; hair-directed therapy runs in parallel. Co-explanations: Androgenetic / Pattern Hair Loss. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-14 — Acute Telogen Effluvium

Case 14 of 25

Patient Questionnaire

Sex: Female

Age group: young adult

Duration of concern: Less than 3 months

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Normal scalp

Self-reported causes: Recent illness

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Diffuse telogen-phase shedding
- Mild active shedding
- Acute timeline (< 6 months)
- Female
- Young adult (< 25)
- Recent illness (post-illness phase)

Detected Drivers

- Telogen cycle disruption — Primary driver

Diagnosis

Primary diagnosis: Telogen Effluvium

Severity: Mild

Root Cause Explanation

22-year-old female presenting with diffuse synchronized shedding lagging a clear precipitant (post-fever) by the expected 2–4 month telogen cycle. Shedding intensity (100–200 hairs/day), acute duration, and absence of pattern topology produce a dominant telogen-cycle-disruption activation. Architectural follicle integrity is preserved — this is cycle-timing pathology, fully reversible, and protocol class TE_ACUTE applies with cycle-restoration and stress-downregulation therapy needs.

Biological Mechanism

Stress-axis activation accelerates anagen-to-telogen transition. A wave of synchronized follicles enters telogen simultaneously, shedding the dormant club hairs approximately 2–4 months after the precipitant. The follicle bulb is intact; growth resumes spontaneously once the precipitant resolves.

Protocol Recommendation

- Hair-cycle restoration
- Stress-axis downregulation
- Nutritional repletion

Cycle-restoration protocol with stress-axis downregulation and pre-emptive substrate repletion to support the rebound anagen wave. Reversibility is the dominant message; no permanent loss is expected from the current episode. Therapy axes engaged: Hair-cycle restoration, Stress-axis downregulation, Nutritional repletion.

Monitoring Plan

Required

- Shed count — monthly diary
- Global photography — every 3 months

Recommended

- Stress / PHQ-9 — monthly

Expected Outcomes

Active shedding subsides within 8–16 weeks. New growth visible from 12–16 weeks. Full visual recovery generally at 6–9 months, longer for chronic phenotypes.

Patient Narrative

A stress event 2–4 months ago pushed many hairs into a resting phase, which now shows up as visible shedding. This is fully reversible. Most regrowth happens over the next 4–6 months as the cycle resets.

Doctor Narrative

Diffuse synchronized telogen release with an identifiable precipitant. Follicular architecture preserved; the process is fully reversible. Protocol favours cycle restoration with stress-axis downregulation. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-15 — Acute Telogen Effluvium

Case 15 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: 3–6 months

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Normal scalp

Self-reported causes: Postpartum

Lifestyle factors: None

Thyroid: None

Hormonal state: Postpartum

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Diffuse telogen-phase shedding
- Heavy active shedding
- Acute timeline (< 6 months)
- Female
- Adult (25–39)
- Postpartum, lactating

Detected Drivers

- Telogen cycle disruption — Primary driver
- Endocrine dysregulation — Supporting driver

Diagnosis

Primary diagnosis: Telogen Effluvium

Severity: Moderate

Co-explanations: Endocrine-Driven Hair Loss

Root Cause Explanation

26-year-old female presenting with diffuse synchronized shedding lagging a clear precipitant (postpartum-lactating) by the expected 2–4 month telogen cycle. Shedding intensity (100–200 hairs/day), acute duration, and absence of pattern topology produce a dominant telogen-cycle-disruption activation. Architectural follicle integrity is preserved — this is cycle-timing pathology, fully reversible, and protocol class TE_ACUTE applies with cycle-restoration and stress-downregulation therapy needs. Postpartum hormonal shift co-explains and is acknowledged in the narrative.

Biological Mechanism

Stress-axis activation accelerates anagen-to-telogen transition. A wave of synchronized follicles enters

telogen simultaneously, shedding the dormant club hairs approximately 2–4 months after the precipitant. The follicle bulb is intact; growth resumes spontaneously once the precipitant resolves.

Protocol Recommendation

- Hair-cycle restoration
- Stress-axis downregulation
- Nutritional repletion

Cycle-restoration protocol with stress-axis downregulation and pre-emptive substrate repletion to support the rebound anagen wave. Reversibility is the dominant message; no permanent loss is expected from the current episode. Therapy axes engaged: Hair-cycle restoration, Stress-axis downregulation, Nutritional repletion.

Monitoring Plan

Required

- Shed count — monthly diary
- Global photography — every 3 months
- TSH — repeat at 3 months

Recommended

- Stress / PHQ-9 — monthly

Expected Outcomes

Active shedding subsides within 8–16 weeks. New growth visible from 12–16 weeks. Full visual recovery generally at 6–9 months, longer for chronic phenotypes.

Patient Narrative

A stress event 2–4 months ago pushed many hairs into a resting phase, which now shows up as visible shedding. This is fully reversible. Most regrowth happens over the next 4–6 months as the cycle resets. We are also addressing the contributing role of Endocrine-Driven Hair Loss.

Doctor Narrative

Diffuse synchronized telogen release with an identifiable precipitant. Follicular architecture preserved; the process is fully reversible. Protocol favours cycle restoration with stress-axis downregulation. Co-explanations: Endocrine-Driven Hair Loss.

Case RC-16 — Acute Telogen Effluvium

Case 16 of 25

Patient Questionnaire

Sex: Male

Age group: adult

Duration of concern: 3–6 months

Shedding pattern: more than 200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Normal scalp

Self-reported causes: Crash diet

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: Iron deficiency

Gut symptoms: None

Diet pattern: Crash dieting

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Diffuse telogen-phase shedding
- Heavy active shedding
- Acute timeline (< 6 months)
- Male
- Adult (25–39)
- Crash dieting / rapid weight loss
- Iron / ferritin deficiency reported

Detected Drivers

- Telogen cycle disruption — Primary driver
- Nutritional substrate limitation — Supporting driver

Diagnosis

Primary diagnosis: Telogen Effluvium

Severity: Severe

Co-explanations: Nutritional Hair Loss

Root Cause Explanation

36-year-old male presenting with diffuse synchronized shedding lagging a clear precipitant (crash-diet) by the expected 2–4 month telogen cycle. Shedding intensity (more than 200 hairs/day), acute duration, and absence of pattern topology produce a dominant telogen-cycle-disruption activation. Architectural follicle integrity is preserved — this is cycle-timing pathology, fully reversible, and protocol class TE_ACUTE applies with cycle-restoration and stress-downregulation therapy needs. Nutritional-limitation co-activates and is repleted in parallel as co-explanation.

Biological Mechanism

Stress-axis activation accelerates anagen-to-telogen transition. A wave of synchronized follicles enters telogen simultaneously, shedding the dormant club hairs approximately 2–4 months after the precipitant. The follicle bulb is intact; growth resumes spontaneously once the precipitant resolves.

Protocol Recommendation

- Hair-cycle restoration
- Stress-axis downregulation
- Nutritional repletion

Cycle-restoration protocol with stress-axis downregulation and pre-emptive substrate repletion to support the rebound anagen wave. Reversibility is the dominant message; no permanent loss is expected from the current episode. Therapy axes engaged: Hair-cycle restoration, Stress-axis downregulation, Nutritional repletion.

Monitoring Plan

Required

- Shed count — monthly diary
- Global photography — every 3 months
- Ferritin — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months

Recommended

- Stress / PHQ-9 — monthly

Expected Outcomes

Active shedding subsides within 8–16 weeks. New growth visible from 12–16 weeks. Full visual recovery generally at 6–9 months, longer for chronic phenotypes.

Patient Narrative

A stress event 2–4 months ago pushed many hairs into a resting phase, which now shows up as visible shedding. This is fully reversible. Most regrowth happens over the next 4–6 months as the cycle resets. We are also addressing the contributing role of Nutritional Hair Loss.

Doctor Narrative

Diffuse synchronized telogen release with an identifiable precipitant. Follicular architecture preserved; the process is fully reversible. Protocol favours cycle restoration with stress-axis downregulation. Co-explanations: Nutritional Hair Loss.

Case RC-17 — Chronic Telogen Effluvium

Case 17 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Normal scalp

Self-reported causes: Stress

Lifestyle factors: Chronic stress; Poor sleep

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Diffuse telogen-phase shedding
- Chronic timeline (> 12 months)
- Mild active shedding
- Female
- Adult (25–39)
- Chronic stress / poor sleep

Detected Drivers

- Telogen cycle disruption — Primary driver

Diagnosis

Primary diagnosis: Telogen Effluvium

Severity: Mild

Root Cause Explanation

28-year-old female with persistent diffuse shedding > 12 months, no architectural pattern change, documented chronic stress load. Chronic-duration marker plus diffuse shedding without pattern topology produces telogen-cycle-disruption leadership at modest but sustained activation. Reversibility is partial-to-full once the chronic driver is addressed; protocol class TE_CHRONIC mandates comprehensive deficiency workup in monitoring.

Biological Mechanism

Stress-axis activation accelerates anagen-to-telogen transition. A wave of synchronized follicles enters telogen simultaneously, shedding the dormant club hairs approximately 2–4 months after the precipitant. The follicle bulb is intact; growth resumes spontaneously once the precipitant resolves.

Protocol Recommendation

- Hair-cycle restoration
- Stress-axis downregulation
- Nutritional repletion

Cycle-restoration protocol with stress-axis downregulation and pre-emptive substrate repletion to support the rebound anagen wave. Reversibility is the dominant message; no permanent loss is expected from the current episode. Therapy axes engaged: Hair-cycle restoration, Stress-axis downregulation, Nutritional repletion.

Monitoring Plan

Required

- Shed count — monthly diary
- Global photography — every 3 months
- Ferritin — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months
- TSH — repeat at 3 months

Recommended

- Stress / PHQ-9 — monthly

Expected Outcomes

Active shedding subsides within 8–16 weeks. New growth visible from 12–16 weeks. Full visual recovery generally at 6–9 months, longer for chronic phenotypes.

Patient Narrative

A stress event 2–4 months ago pushed many hairs into a resting phase, which now shows up as visible shedding. This is fully reversible. Most regrowth happens over the next 4–6 months as the cycle resets.

Doctor Narrative

Diffuse synchronized telogen release with an identifiable precipitant. Follicular architecture preserved; the process is fully reversible. Protocol favours cycle restoration with stress-axis downregulation. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-18 — Chronic Telogen Effluvium

Case 18 of 25

Patient Questionnaire

Sex: Male

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Normal scalp

Self-reported causes: Chronic medical condition

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Diffuse telogen-phase shedding
- Chronic timeline (> 12 months)
- Mild active shedding
- Male
- Adult (25–39)
- Chronic medical condition on medication

Detected Drivers

- Telogen cycle disruption — Primary driver

Diagnosis

Primary diagnosis: Telogen Effluvium

Severity: Moderate

Root Cause Explanation

32-year-old male with persistent diffuse shedding > 12 months, no architectural pattern change, no identifiable single precipitant despite thorough workup. Chronic-duration marker plus diffuse shedding without pattern topology produces telogen-cycle-disruption leadership at modest but sustained activation. Reversibility is partial-to-full once the chronic driver is addressed; protocol class TE_CHRONIC mandates comprehensive deficiency workup in monitoring.

Biological Mechanism

Stress-axis activation accelerates anagen-to-telogen transition. A wave of synchronized follicles enters telogen simultaneously, shedding the dormant club hairs approximately 2–4 months after the precipitant. The follicle bulb is intact; growth resumes spontaneously once the precipitant resolves.

Protocol Recommendation

- Hair-cycle restoration
- Stress-axis downregulation
- Nutritional repletion

Cycle-restoration protocol with stress-axis downregulation and pre-emptive substrate repletion to support the rebound anagen wave. Reversibility is the dominant message; no permanent loss is expected from the current episode. Therapy axes engaged: Hair-cycle restoration, Stress-axis downregulation, Nutritional repletion.

Monitoring Plan

Required

- Shed count — monthly diary
- Global photography — every 3 months
- Ferritin — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months
- TSH — repeat at 3 months

Recommended

- Stress / PHQ-9 — monthly

Expected Outcomes

Active shedding subsides within 8–16 weeks. New growth visible from 12–16 weeks. Full visual recovery generally at 6–9 months, longer for chronic phenotypes.

Patient Narrative

A stress event 2–4 months ago pushed many hairs into a resting phase, which now shows up as visible shedding. This is fully reversible. Most regrowth happens over the next 4–6 months as the cycle resets.

Doctor Narrative

Diffuse synchronized telogen release with an identifiable precipitant. Follicular architecture preserved; the process is fully reversible. Protocol favours cycle restoration with stress-axis downregulation.

Case RC-19 — Alopecia Areata Spectrum

Case 19 of 25

Patient Questionnaire

Sex: Female

Age group: young adult

Duration of concern: Less than 3 months

Shedding pattern: patches without diffuse fall

Reported grade: Grade 1

Hair changes: Patchy bald spots

Scalp state: Normal scalp

Self-reported causes: Autoimmune

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: Alopecia areata history

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Patchy, well-demarcated loss
- Alopecia areata history
- Female
- Young adult (< 25)

Detected Drivers

- Immune dysregulation — Primary driver

Diagnosis

Primary diagnosis: Autoimmune Alopecia (Alopecia Areata spectrum)

Severity: Mild

Root Cause Explanation

12-year-old female with well-demarcated patchy loss and confirmed alopecia areata history. The patchy-loss-marker and alopecia-areata-history signals together carry the strongest LLRs in the autoimmune cause definition; immune-dysregulation pathway leads at "e0.60 activation. Protocol class is AUTOIMMUNE_AA — treatment focuses on immune quiescence and expectation setting for relapse/remission rather than miniaturization-style intervention.

Biological Mechanism

Loss of immune privilege at the follicle bulge permits CD8+ T-cell infiltration, producing the alopecia areata phenotype with well-demarcated loss. Scarring forms reflect irreversible perifollicular fibrosis. Treatment targets immune quiescence; the course is relapsing-remitting in many patients.

Protocol Recommendation

- Autoimmune quiescence

- Immune modulation
- Scalp inflammation control

Autoimmune quiescence protocol with realistic expectation setting; treatment goals are stabilisation and regrowth windows rather than guaranteed restoration. Therapy axes engaged: Autoimmune quiescence, Immune modulation, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Scalp clinical exam — every 3 months

Recommended

- Trichoscopy — every 6 months

Expected Outcomes

Stabilisation is the primary near-term goal. Regrowth windows are variable and patient-specific; relapse cycles are expected and do not constitute treatment failure.

Patient Narrative

Your immune system is reacting against the follicles. Treatment focuses on calming that response. Outcomes vary; we plan for relapse cycles and treat stabilisation as a success outcome.

Doctor Narrative

Immune-mediated follicular targeting. Set relapse-remission expectations explicitly with the patient; outcome metrics track stabilisation rather than guaranteed restoration. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-20 — Alopecia Areata Spectrum

Case 20 of 25

Patient Questionnaire

Sex: Male

Age group: young adult

Duration of concern: Less than 3 months

Shedding pattern: patches without diffuse fall

Reported grade: Grade 1

Hair changes: Patchy bald spots

Scalp state: Normal scalp

Self-reported causes: Autoimmune

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: Alopecia areata history

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Patchy, well-demarcated loss
- Alopecia areata history
- Male
- Young adult (< 25)

Detected Drivers

- Immune dysregulation — Primary driver

Diagnosis

Primary diagnosis: Autoimmune Alopecia (Alopecia Areata spectrum)

Severity: Moderate

Root Cause Explanation

18-year-old male with well-demarcated patchy loss and confirmed alopecia areata history. The patchy-loss-marker and alopecia-areata-history signals together carry the strongest LLRs in the autoimmune cause definition; immune-dysregulation pathway leads at "e0.60 activation. Protocol class is AUTOIMMUNE_AA — treatment focuses on immune quiescence and expectation setting for relapse/remission rather than miniaturization-style intervention.

Biological Mechanism

Loss of immune privilege at the follicle bulge permits CD8+ T-cell infiltration, producing the alopecia areata phenotype with well-demarcated loss. Scarring forms reflect irreversible perifollicular fibrosis. Treatment targets immune quiescence; the course is relapsing-remitting in many patients.

Protocol Recommendation

- Autoimmune quiescence

- Immune modulation
- Scalp inflammation control

Autoimmune quiescence protocol with realistic expectation setting; treatment goals are stabilisation and regrowth windows rather than guaranteed restoration. Therapy axes engaged: Autoimmune quiescence, Immune modulation, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Scalp clinical exam — every 3 months

Recommended

- Trichoscopy — every 6 months

Expected Outcomes

Stabilisation is the primary near-term goal. Regrowth windows are variable and patient-specific; relapse cycles are expected and do not constitute treatment failure.

Patient Narrative

Your immune system is reacting against the follicles. Treatment focuses on calming that response. Outcomes vary; we plan for relapse cycles and treat stabilisation as a success outcome.

Doctor Narrative

Immune-mediated follicular targeting. Set relapse-remission expectations explicitly with the patient; outcome metrics track stabilisation rather than guaranteed restoration.

Case RC-21 — Alopecia Areata Spectrum

Case 21 of 25

Patient Questionnaire

Sex: Female

Age group: young adult

Duration of concern: Less than 3 months

Shedding pattern: patches without diffuse fall

Reported grade: Grade 1

Hair changes: Patchy bald spots

Scalp state: Normal scalp

Self-reported causes: Autoimmune

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: Alopecia areata history

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Patchy, well-demarcated loss
- Alopecia areata history
- Female
- Young adult (< 25)

Detected Drivers

- Immune dysregulation — Primary driver

Diagnosis

Primary diagnosis: Autoimmune Alopecia (Alopecia Areata spectrum)

Severity: Severe

Root Cause Explanation

22-year-old female with well-demarcated patchy loss and confirmed alopecia areata history. The patchy-loss-marker and alopecia-areata-history signals together carry the strongest LLRs in the autoimmune cause definition; immune-dysregulation pathway leads at "e0.60 activation. Protocol class is AUTOIMMUNE_AA — treatment focuses on immune quiescence and expectation setting for relapse/remission rather than miniaturization-style intervention.

Biological Mechanism

Loss of immune privilege at the follicle bulge permits CD8+ T-cell infiltration, producing the alopecia areata phenotype with well-demarcated loss. Scarring forms reflect irreversible perifollicular fibrosis. Treatment targets immune quiescence; the course is relapsing-remitting in many patients.

Protocol Recommendation

- Autoimmune quiescence

- Immune modulation
- Scalp inflammation control

Autoimmune quiescence protocol with realistic expectation setting; treatment goals are stabilisation and regrowth windows rather than guaranteed restoration. Therapy axes engaged: Autoimmune quiescence, Immune modulation, Scalp inflammation control.

Monitoring Plan

Required

- Global photography — every 3 months
- Scalp clinical exam — every 3 months

Recommended

- Trichoscopy — every 6 months

Expected Outcomes

Stabilisation is the primary near-term goal. Regrowth windows are variable and patient-specific; relapse cycles are expected and do not constitute treatment failure.

Patient Narrative

Your immune system is reacting against the follicles. Treatment focuses on calming that response. Outcomes vary; we plan for relapse cycles and treat stabilisation as a success outcome.

Doctor Narrative

Immune-mediated follicular targeting. Set relapse-remission expectations explicitly with the patient; outcome metrics track stabilisation rather than guaranteed restoration.

Case RC-22 — Inflammatory Scalp Disorder

Case 22 of 25

Patient Questionnaire

Sex: Male

Age group: young adult

Duration of concern: 6–12 months

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Dandruff with itching

Self-reported causes: Scalp infection

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Male
- Young adult (< 25)
- Dandruff with itching

Detected Drivers

- Scalp inflammation — Primary driver

Diagnosis

Primary diagnosis: Inflammatory Scalp Disorder

Severity: Mild

Root Cause Explanation

22-year-old male with dominant scalp-surface inflammation signals (seborrheic mild) and mild associated shedding. Scalp-inflammation pathway leads with "e0.60 activation; pattern signal is absent, so the inflammatory cause is the named driver rather than a downstream modifier. Protocol class INFLAMMATORY focuses on scalp decongestion and topical inflammation control; once the surface dermatosis quiets, follicle cycling normalizes.

Biological Mechanism

Malassezia-driven seborrheic inflammation, psoriatic plaques, or perifollicular pustulation disrupt anagen maintenance and produce a pro-shedding micro-environment around the follicle. Calming the scalp surface restores cycling in most cases.

Protocol Recommendation

- Scalp inflammation control
- Scalp decongestion / sebum control

Scalp decongestion and inflammation control; hair density typically follows within 8–12 weeks once the surface dermatosis quiets. Therapy axes engaged: Scalp inflammation control, Scalp decongestion / sebum control.

Monitoring Plan

Required

- Scalp clinical exam — every 3 months
- Global photography — every 3 months

Recommended

- Trichoscopy — every 6 months

Expected Outcomes

Surface dermatosis typically improves within 2–6 weeks. Shedding tied to inflammation subsides over the same window. Density change at 12–16 weeks.

Patient Narrative

The scalp environment around the follicle is inflamed. Calming the scalp is the primary lever. Improvement is generally fast once inflammation settles.

Doctor Narrative

Perifollicular and surface inflammation is the named driver; resolving the scalp-surface dermatosis restores cycling. Re-examine scalp at 6–8 weeks. Presentation is ambiguous — recommend reassessment at 12 weeks before therapy intensification.

Case RC-23 — Inflammatory Scalp Disorder

Case 23 of 25

Patient Questionnaire

Sex: Female

Age group: adult

Duration of concern: 6–12 months

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 1

Hair changes: Diffuse shedding

Scalp state: Oily scalp; Dandruff with itching; Scalp redness

Self-reported causes: Scalp infection

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall

Clinical Findings

- Female
- Adult (25–39)
- Dandruff with itching
- Scalp redness
- Oily scalp

Detected Drivers

- Scalp inflammation — Primary driver

Diagnosis

Primary diagnosis: Inflammatory Scalp Disorder

Severity: Moderate

Root Cause Explanation

28-year-old female with dominant scalp-surface inflammation signals (seborrheic severe) and mild associated shedding. Scalp-inflammation pathway leads with "e0.60 activation; pattern signal is absent, so the inflammatory cause is the named driver rather than a downstream modifier. Protocol class INFLAMMATORY focuses on scalp decongestion and topical inflammation control; once the surface dermatosis quiets, follicle cycling normalizes.

Biological Mechanism

Malassezia-driven seborrheic inflammation, psoriatic plaques, or perifollicular pustulation disrupt anagen maintenance and produce a pro-shedding micro-environment around the follicle. Calming the scalp surface restores cycling in most cases.

Protocol Recommendation

- Scalp inflammation control
- Scalp decongestion / sebum control

Scalp decongestion and inflammation control; hair density typically follows within 8–12 weeks once the surface dermatosis quiets. Therapy axes engaged: Scalp inflammation control, Scalp decongestion / sebum control.

Monitoring Plan

Required

- Scalp clinical exam — every 3 months
- Global photography — every 3 months

Recommended

- Trichoscopy — every 6 months

Expected Outcomes

Surface dermatosis typically improves within 2–6 weeks. Shedding tied to inflammation subsides over the same window. Density change at 12–16 weeks.

Patient Narrative

The scalp environment around the follicle is inflamed. Calming the scalp is the primary lever. Improvement is generally fast once inflammation settles.

Doctor Narrative

Perifollicular and surface inflammation is the named driver; resolving the scalp-surface dermatosis restores cycling. Re-examine scalp at 6–8 weeks.

Case RC-24 — Multifactorial Hair Loss

Case 24 of 25

Patient Questionnaire

Sex: Male

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 3

Hair changes: Thinning at crown; Diffuse shedding

Scalp state: Normal scalp

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Diffuse telogen-phase shedding
- Early-grade severity (1–3)
- Chronic timeline (> 12 months)
- Male
- Adult (25–39)
- Family history of hair loss

Detected Drivers

- Follicular miniaturization — Supporting driver
- Telogen cycle disruption — Supporting driver

Diagnosis

Primary diagnosis: Multifactorial Hair Loss

Severity: Moderate

Co-explanations: Androgenetic / Pattern Hair Loss; Telogen Effluvium

Root Cause Explanation

28-year-old male with simultaneously activated miniaturization, telogen-cycle-disruption, and a third driver pathway (aga te overlap). The compositeRule in the cause registry ("e3 pathways above 0.40 AND single-cause dissent gap < 0.06) is the only path by which multifactorial-hair-loss promotes to lead — naive Bayes would surface AGA or TE alone, missing the composite reality. Protocol class MULTIFACTORIAL coordinates therapy combinatorics across DHT suppression, cycle restoration, inflammation control, and substrate repletion.

Biological Mechanism

Multiple converging mechanisms — typically miniaturization, telogen-cycle disruption, and an endocrine/inflammatory/nutritional axis — are simultaneously active above clinical threshold. No single mechanism dominates; treatment must address the top contributors in parallel for meaningful improvement.

Protocol Recommendation

- DHT suppression
- Hair-cycle restoration
- Scalp inflammation control
- Nutritional repletion
- Endocrine optimisation

Multifactorial coordination protocol — DHT suppression, cycle restoration, scalp control, and substrate repletion run concurrently with staged reassessment. Therapy axes engaged: DHT suppression, Hair-cycle restoration, Scalp inflammation control, Nutritional repletion, Endocrine optimisation.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months
- Ferritin — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months
- TSH — repeat at 3 months

Recommended

- Shed count — monthly diary
- Stress / PHQ-9 — monthly

Expected Outcomes

Recovery trajectory is composite and slower than any single-cause case. Realistic milestones at 4, 6, and 9 months across the engaged axes.

Patient Narrative

Multiple causes are working together here. We need to address more than one driver to see meaningful improvement — single-lever fixes will under-deliver. We are also addressing the contributing role of Androgenetic / Pattern Hair Loss and Telogen Effluvium.

Doctor Narrative

No single driver dominates; the case requires a composite explanation. Treatment addresses the top co-leading causes in parallel with staged reassessment. Co-explanations: Androgenetic / Pattern Hair Loss, Telogen Effluvium. Conflicting signals present — multifactorial framing preferred.

Case RC-25 — Multifactorial Hair Loss

Case 25 of 25

Patient Questionnaire

Sex: Male

Age group: adult

Duration of concern: More than 1 year

Shedding pattern: 100–200 hairs/day

Reported grade: Grade 4

Hair changes: Thinning at crown; Diffuse shedding

Scalp state: Oily scalp; Dandruff with itching

Self-reported causes: Family history

Lifestyle factors: None

Thyroid: None

Hormonal state: None

Immunity / autoimmune: None

Reported deficiencies: None reported

Gut symptoms: None

Diet pattern: Mixed

Current treatments: None

Treatment goals: Reduce hair fall; Regrow lost hair

Clinical Findings

- Pattern thinning / widening parting
- Diffuse telogen-phase shedding
- Advanced-grade severity (4–5)
- Chronic timeline (> 12 months)
- Male
- Adult (25–39)
- Family history of hair loss
- Dandruff with itching
- Oily scalp

Detected Drivers

- Follicular miniaturization — Supporting driver
- Telogen cycle disruption — Supporting driver
- Scalp inflammation — Supporting driver

Diagnosis

Primary diagnosis: Multifactorial Hair Loss

Severity: Severe

Co-explanations: Androgenetic / Pattern Hair Loss; Telogen Effluvium

Root Cause Explanation

32-year-old male with simultaneously activated miniaturization, telogen-cycle-disruption, and a third driver pathway (aga inflam overlap). The compositeRule in the cause registry ("e3 pathways above 0.40 AND single-cause dissent gap < 0.06) is the only path by which multifactorial-hair-loss promotes to lead — naive Bayes would surface AGA or TE alone, missing the composite reality. Protocol class MULTIFACTORIAL coordinates therapy combinatorics across DHT suppression, cycle restoration,

inflammation control, and substrate repletion.

Biological Mechanism

Multiple converging mechanisms — typically miniaturization, telogen-cycle disruption, and an endocrine/inflammatory/nutritional axis — are simultaneously active above clinical threshold. No single mechanism dominates; treatment must address the top contributors in parallel for meaningful improvement.

Protocol Recommendation

- DHT suppression
- Hair-cycle restoration
- Scalp inflammation control
- Nutritional repletion
- Endocrine optimisation

Multifactorial coordination protocol — DHT suppression, cycle restoration, scalp control, and substrate repletion run concurrently with staged reassessment. Therapy axes engaged: DHT suppression, Hair-cycle restoration, Scalp inflammation control, Nutritional repletion, Endocrine optimisation.

Monitoring Plan

Required

- Global photography — every 3 months
- Trichoscopy — every 6 months
- Ferritin — repeat at 4 months
- Vitamin D (25-OH) — repeat at 4 months
- TSH — repeat at 3 months

Recommended

- Shed count — monthly diary
- Stress / PHQ-9 — monthly

Expected Outcomes

Recovery trajectory is composite and slower than any single-cause case. Realistic milestones at 4, 6, and 9 months across the engaged axes.

Patient Narrative

Multiple causes are working together here. We need to address more than one driver to see meaningful improvement — single-lever fixes will under-deliver. We are also addressing the contributing role of Androgenetic / Pattern Hair Loss and Telogen Effluvium.

Doctor Narrative

No single driver dominates; the case requires a composite explanation. Treatment addresses the top co-leading causes in parallel with staged reassessment. Co-explanations: Androgenetic / Pattern Hair Loss, Telogen Effluvium. Conflicting signals present — multifactorial framing preferred.

